Hypothesis Testing

CS 215: Fall 2025
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Introduction: Hypothesis testing

- A **statistical hypothesis test** = statement about a set of parameters regarding the distribution of a population, usually tested using only some *n* samples from that distribution
- **Example 1**
  - Suppose a construction firm has purchased a large number of cables. The manufacturer claims that the average breaking strength of the cables is 7000 PSI (pounds per square inch).
  - The firm decides to verify the manufacturer's claim. The firm tests say 15 of these cables to determine their breaking strength
  - The results of these tests will be used to determine whether or not they accept the cable manufacturer's hypothesis that the population mean breaking strength is at least 7000 PSI.
- **Example 2**
  - Consider samples from N(θ,1) where mean θ is unknown
  - Given samples from this distribution, we want to determine: Is $\theta < 1$?

Introduction: Hypothesis testing

- In example 2, suppose the samples average is 1.25 in one scenario and 5.25 in another scenario.
- We intuitively feel that it is less likely for the samples to be distributed from N(θ,1) for $\theta < 1$ in the second scenario as compared to the first.
- If the random samples are deemed to be consistent with the hypothesis under consideration, we say that the hypothesis has been **accepted**.
- Otherwise, we say the hypothesis has been **rejected**.
- How do we judge the consistency?
- We could compute the sample mean X' and then put some threshold on |θ-X'|, but such thresholds do not make use of the variance information properly.

Concept of null hypothesis

- Consider a population with distribution $F(\theta)$ where θ is unknown.
- Let us say we want to test a certain hypothesis H_0 regarding θ.
- For this, we are given samples $X_1, X_2, \ldots, X_n$ and based on them we decide whether or not H_0 is true.
- H_0 is often called the **null hypothesis**.
- For example, the null hypothesis could be $\theta = 1$ or $\theta < 1$ or $\theta > 1$.
- $\theta = 1$ is called a **simple hypothesis**: if true, it completely specifies the population distribution
- The other two are called **compound hypotheses**: if true, they do not completely specify the population distribution

Concept of Type I and Type II errors

- For example if X_1, X_2,...,X_n are drawn from $N(\theta,1)$.
- Then we could say that the null hypothesis H_0: $\theta = 1$ is to be rejected if

$$|\bar{X} - 1| > 1.96/\sqrt{n}$$

This is a standard normal random variable

- During these tests, there can be two types of errors:
  - **Type I error:** the test incorrectly recommends rejection of H_0 (even if it is true)
  - **Type II error:** the test incorrectly recommends acceptance of H_0 (even if it is false)
- Ideally, H_0 should be rejected only if the samples are very unlikely when H_0 is true.
- We design the test in such a way that whenever H_0 is true, the probability of rejection based on the samples (type I error) is never greater than α, called the **significance level** of the test.
- It is a parameter set in advance. Typical values of α are 0.05 or 0.01.

Concept of critical region

- The **critical region** C is a region of the sample space. The null hypothesis H0 is to be **rejected** if the samples $X_1,\ldots,X_n$ **belong to C**, and **accepted** if they do not **belong to C**.
- In the earlier example, the critical region is given by

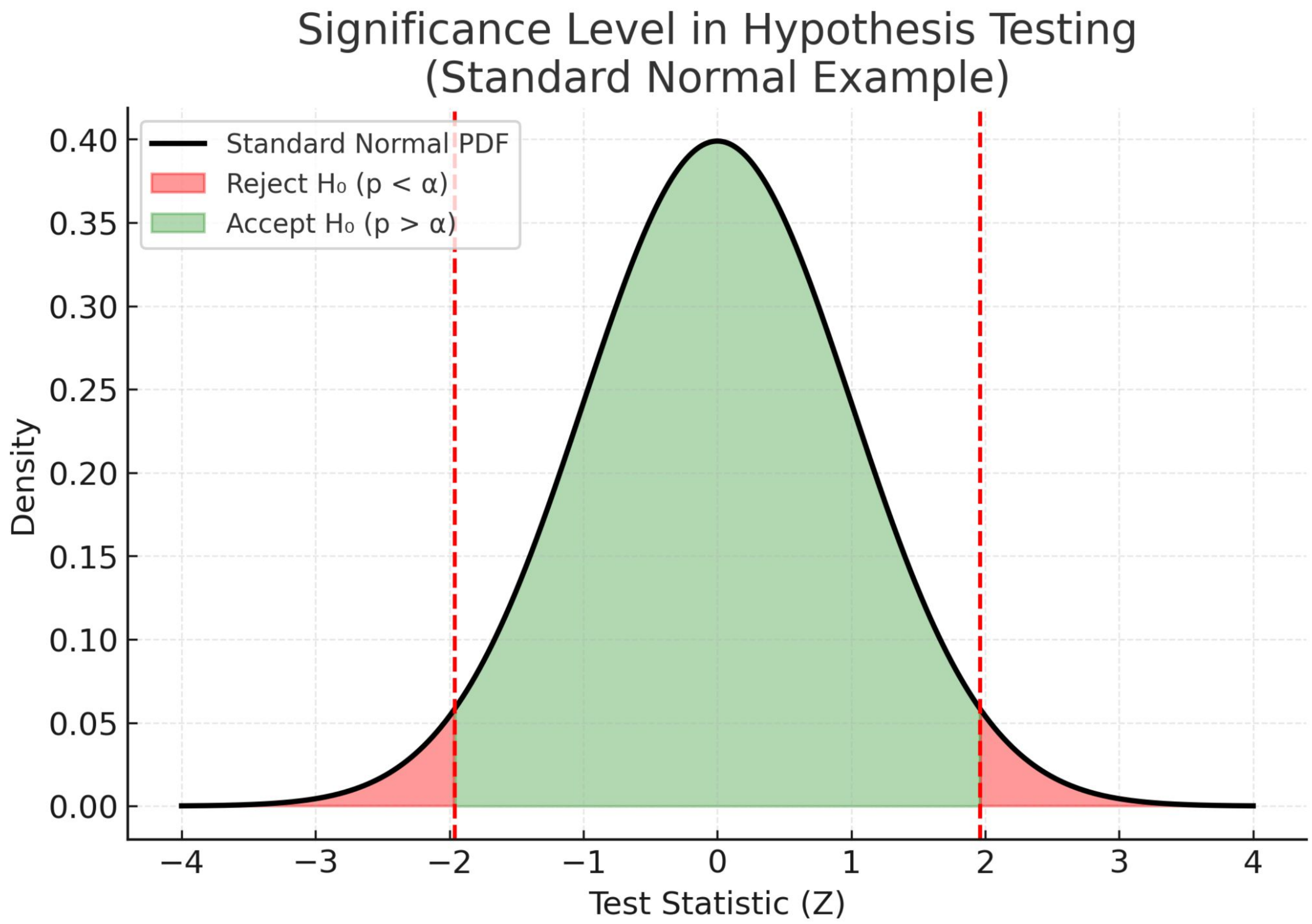
$$C = \{(X_1, \ldots, X_n) : \ |\bar{X} - 1| > 1.96/\sqrt{n}\}$$

Concept of p-value

- Consider the hypothesis H0

  $$H_0 : \theta \in w$$

  - We use a point estimator d(X) of θ given samples $X_1, X_2, \ldots, X_n$.
  - d(X) is sometimes called the **test statistic**.
  - If d(X) is far away from *w*, we **reject** the null hypothesis; otherwise, we **accept** the null hypothesis.
  - In general, we compute the probability that θ would be at least as extreme as the value d(X). This is called the **p-value** of the test.
  - If the p-value **exceeds** the significance level, the null hypothesis is **accepted** – see next slide for an illustration.
  - If the p-value is **less than** the significance level, the null hypothesis is **rejected** – see next slide for an illustration.



Significance Level in Hypothesis Testing
(Standard Normal Example)
Standard Normal PDF
Reject H₀ (p < α)
Accept H₀ (p > α)
Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4
Test Statistic (Z)

Hypothesis Tests

- We will develop hypothesis tests for:
  a. Mean of a normal distribution with known variance (**z-test**)
  b. Mean of a normal distribution with unknown variance (**t-test**)
  c. Testing the equality of the means of two normal populations with known variance (**paired z-test**)
  d. Testing the equality of the means of two normal populations with unknown variance (**paired t-test**)
  e. Hypothesis testing for the variance of a normal population (**chi-square test**)
  f. Mean of Bernoulli populations
  g. Equality of the means of two Bernoulli populations (**Fischer Iwrin test**)

(a) Hypothesis test for mean of a normal distribution with known variance

- The hypothesis to be tested is $H_0 : \mu = \mu_0$
- The alternative hypothesis is $H_1 : \mu \neq \mu_0$
- The point estimator for μ is given by $\overline{X} = \sum_{i=1}^{n} X_i/n$
- It is reasonable to accept H_0 if bar(X) is not "too far" from μ_0. That is, the critical region of the test would be of the form

$$C = \{X_1, \ldots, X_n : |\overline{X} - \mu_0| > c\}$$

- We need to decide the value of c above for which the probability of type I error equals α, i.e. we decide c such that

$$P_{\mu_0}\{|\overline{X} - \mu_0| > c\} = \alpha$$

This is the probability computed assuming that $\mu = \mu_0$

(a) Hypothesis test for mean of a normal distribution with known variance

- When $\mu = \mu_0$, then bar(X) is normally distributed with mean μ and variance σ^2/n.
- The following random variable Z ~ N(0,1)

$$Z \equiv \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

This is the probability computed assuming that μ=μ_0

- So we have

$$P_{\mu_0}\{|\overline{X} - \mu_0| > c\} = \alpha \longrightarrow P\left\{|Z| > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

$$2P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha \longrightarrow P\{Z > z_{\alpha/2}\} = \alpha/2 \longrightarrow c = \frac{z_{\alpha/2}\sigma}{\sqrt{n}}$$

(a) Hypothesis test for mean of a normal distribution with known variance

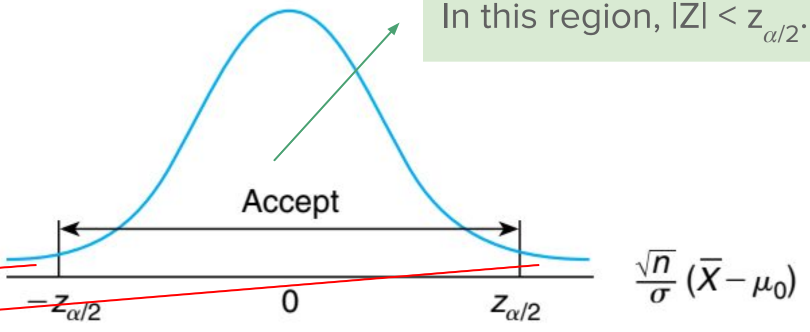
- The final hypothesis test is given by

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| > z_{\alpha/2}$$

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| \leq z_{\alpha/2}$$

- This is also called a **z-test**.

For chosen significance level α, we have the threshold $z_{\alpha/2}$. In this region, $|Z| < z_{\alpha/2}$.



For chosen significance level α, we have the threshold $z_{\alpha/2}$. In this region, $|Z| > z_{\alpha/2}$.

(a) Clarification: Hypothesis test for mean of a normal distribution with known variance

- As given in the earlier slides, the test statistic is Z and is computed from samples X1, X2,...,Xn (note: we know that these samples are from a Gaussian distribution with standard deviation σ).
- The null hypothesis is $\mu = \mu_0$ (where μ_0 is known).
- The significance level α is chosen by us. For example, if we choose α = 0.05, then $z_{\alpha/2}$ = 1.96 (that is $1-\Phi(1.96) = 0.025 = \alpha/2$ where $\Phi(.)$ is the standard normal CDF).
- If Z = 1.68 in some experiment then :
  - Since $|Z| < z_{\alpha/2}$ (1.68 < 1.96), we accept the null hypothesis at significance level α = 0.05.
  - Another way to look at this:
    - The p-value = $P(|Z| >= 1.68)$ assuming that the samples are from $N(\mu_0,\sigma^2)$.
    - $P(|Z| >= 1.68) = 2(1-0.9535) = 0.093$.
    - As 0.093 (p-value) > 0.05 (chosen significance level), we accept the null hypothesis at 0.05 significance level – or any significance level **less than** 0.093

(a) Clarification: Hypothesis test for mean of a normal distribution with known variance

- If we had chosen α = 0.1, then $z_{\alpha/2}$ = 1.645.
- If Z = 1.68 in the same experiment then :
  - Since $Z > z_{\alpha/2}$ (1.68 > 1.645), we reject the null hypothesis at the significance level α = 0.1
  - Another way to interpret this:
    - The p-value = P(|Z| >= 1.68) assuming that the samples are from $N(\mu_0,\sigma^2)$.
    - P(|Z| >= 1.68) = 2(1-0.9535) = 0.093.
    - As 0.093 (p-value) < 0.1 (chosen significance level), we reject the null hypothesis at 0.1 significance level – note that we would still accept the null hypothesis at 0.05 significance.
    - **A claim that you accepted the null hypothesis at 10% significance level is a stronger claim than one where you accepted the null hypothesis at 5% significance level** – why?

Table 1: Table of the Standard Normal Cumulative Distribution Function $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Link

(a) Hypothesis test for mean of a normal distribution with known variance

- Now, we will study the type-II error of this test.
- That is: what is the probability of accepting the null hypothesis H_0 when the true mean μ is **unequal** to μ_0?
- We define the following probability:

$$
\begin{aligned}
\beta(\mu) &= P_\mu\{\text{acceptance of } H_0\} \\
&= P_\mu\left\{\left|\frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}\right| \le z_{\alpha/2}\right\} \\
&= P_\mu\left\{-z_{\alpha/2} \le \frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right\}
\end{aligned}
$$

This represents the probability that H0 is accepted when the true mean is μ. It is called the **operating characteristic (OC) curve**.

$$Z \equiv \frac{\overline{X} - \mu}{\sigma / \sqrt{n}} \sim \mathcal{N}(0, 1) \qquad \beta(\mu) = P_\mu \left\{ -z_{\alpha/2} \leq \frac{\overline{X} - \mu_0}{\sigma / \sqrt{n}} \leq z_{\alpha/2} \right\}$$

$$= P_\mu \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq \frac{\overline{X} - \mu_0 - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\}$$

$$= P_\mu \left\{ -z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \leq Z - \frac{\mu_0}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\mu}{\sigma/\sqrt{n}} \right\}$$

$$= P \left\{ \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \leq Z \leq \frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right\}$$

$$= \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} + z_{\alpha/2} \right) - \Phi\left(\frac{\mu_0 - \mu}{\sigma/\sqrt{n}} - z_{\alpha/2} \right)$$

The probability of accepting the null hypothesis H_0 drops sharply as the difference between μ and μ_0 increases.

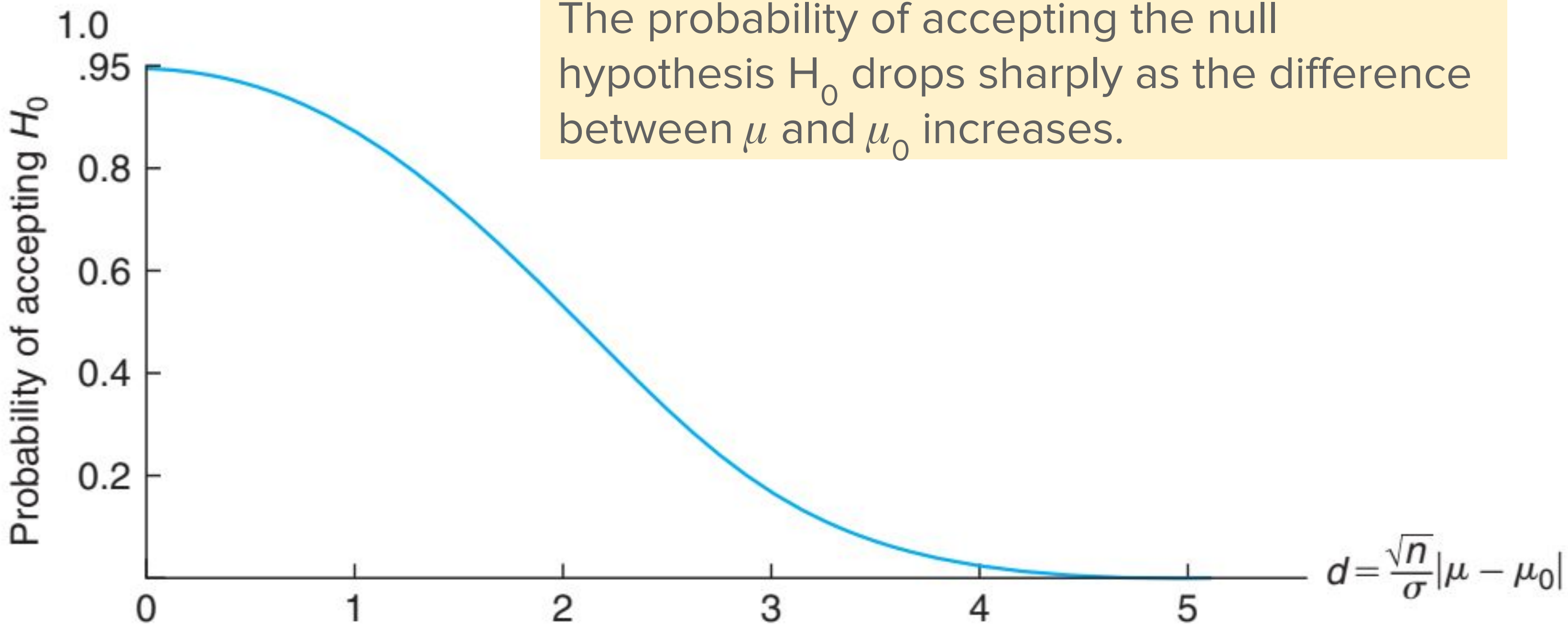


FIGURE 8.2 *The OC curve for the two-sided normal test for significance level* $\alpha = .05$.

1-$\beta(\mu)$ is called the **power function** of the test. It represents the probability of rejection of H_0 when the mean is μ.

- What is the sample size n such that the probability of accepting H_0: $\mu=\mu_0$, when the mean is actually μ_1, is equal to some β?
- That is we want to find *n* for which $\beta(\mu_1)$ = some β.

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}+z_{\alpha/2}\right)-\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\approx\beta$$

- Solving this for *n* in closed form is difficult, so we follow the steps below for the case $\mu_1 > \mu_0$.
- A similar set of steps can be worked out for $\mu_0 < \mu_1$.

$$\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\leq -z_{\alpha/2}$$

As $\boldsymbol{\Phi}$ is an increasing function

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\leq\Phi(-z_{\alpha/2})=P\{Z\leq -z_{\alpha/2}\}=P\{Z\geq z_{\alpha/2}\}=\alpha/2$$

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\leq\Phi(-z_{\alpha/2})=P\{Z\leq -z_{\alpha/2}\}=P\{Z\geq z_{\alpha/2}\}=\alpha/2$$

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\approx 0$$

A reasonable approx. as μ0-μ1 < 0 and so $\boldsymbol{\Phi}$ will be very small

$$\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}+z_{\alpha/2}\right)-\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}-z_{\alpha/2}\right)\approx\beta$$

$$\beta\approx\Phi\left(\frac{\sqrt{n}(\mu_0-\mu_1)}{\sigma}+z_{\alpha/2}\right)$$

$$\beta=P\{Z>z_\beta\}=P\{Z<-z_\beta\}=\Phi(-z_\beta)$$

$$-z_\beta\approx(\mu_0-\mu_1)\frac{\sqrt{n}}{\sigma}+z_{\alpha/2}$$

$$n\approx\frac{(z_{\alpha/2}+z_\beta)^2\sigma^2}{(\mu_1-\mu_0)^2}$$

Example

- A signal value μ is being transmitted from location A to B.
- Suppose the value observed at B is distributed as N(μ,4).
- Test the hypothesis at significance level 5%, that μ = 8 if the average value of the observations at B over n = 5 samples is 9.5.
- **Solution:**

$$\frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| = \frac{\sqrt{5}}{2}(1.5) = 1.68$$

$$z_{.025} = 1.96,$$

1.68 < 1.96 so hypothesis is accepted at 5 percent significance level!

The hypothesis would have been rejected at 10 percent significance level as 1.68 > $z_{0.05}$ = 1.645

(a.1) One sided test for mean of a normal distribution known variance

- In the earlier test, we rejected H0 if the empirical mean was very **far** from μ (could be much less than μ or much greater than μ).
- The earlier test was a **two-sided test**.
- Now, we will consider a **one-sided test**.
- Consider the null hypothesis H0: μ <= μ0.
- The alternative hypothesis here is H1: μ > μ0.
- In other words, we will now **not** reject H0 when the empirical mean is much smaller than μ0.
- The new critical region is:

$$C = \{(X_1, \ldots, X_n) : \overline{X} - \mu_0 > c\}$$

(a.1) One sided test for mean of a normal distribution - known variance

- The probability of rejection should equal α when H0 is true, we need the value of c such that

$$P_{\mu_0}\{\overline{X} - \mu_0 > c\} = \alpha$$

- So we have

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma}$$

$$P\left\{Z > \frac{c\sqrt{n}}{\sigma}\right\} = \alpha$$

$$P\{Z > z_\alpha\} = \alpha$$

$$c = \frac{z_\alpha \sigma}{\sqrt{n}}$$

(a.1) One sided test for mean of a normal distribution: known variance

One-sided

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) \leq z_\alpha$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}(\overline{X} - \mu_0) > z_\alpha$$

Two-sided

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| > z_{\alpha/2}$$

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{\sqrt{n}}{\sigma}|\overline{X} - \mu_0| \leq z_{\alpha/2}$$

Summary

- If we have very strong confidence that H0 is actually true, then we should set a low significance level α in our hypothesis test.
- To see this, note again that the p-value must exceed α.
- If α is small, it means that the corresponding statistic value is very far away from the true mean.
- In other words, we will reject the null hypothesis only if the sample mean is truly **very** far away from the true mean.

(b) Hypothesis testing: mean of a normal distribution with unknown variance

- Consider the null and alternative hypotheses as follows:

$$H_0 : \mu = \mu_0 \qquad H_1 : \mu \neq \mu_0$$

- When the variance is known, the hypothesis test had the following form:

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2}$$

- When the variance is not known, the test statistic will have the following form:

$$\left| \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \right| \qquad S^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$$

This statistic is not Gaussian distributed – it is a student-t random variable

The student-t distribution

- If Z has a standard normal distribution and **χ**^2n has a chi-square distribution with *n* degrees of freedom, then the following random variable has a student-t distribution with *n* degrees of freedom (if Z and **χ**^2n are independent):

$$T_n = \frac{Z}{\sqrt{\chi_n^2/n}}$$

$$f_{T_n}(x) = c(1 + x^2/n)^{-(n+1)/2}, c = \frac{1}{\sqrt{n}B(n/2, 1/2)}$$

$$B(x, y) = \int_0^\infty t^{x-1}(1+t)^{-x-y}dt$$

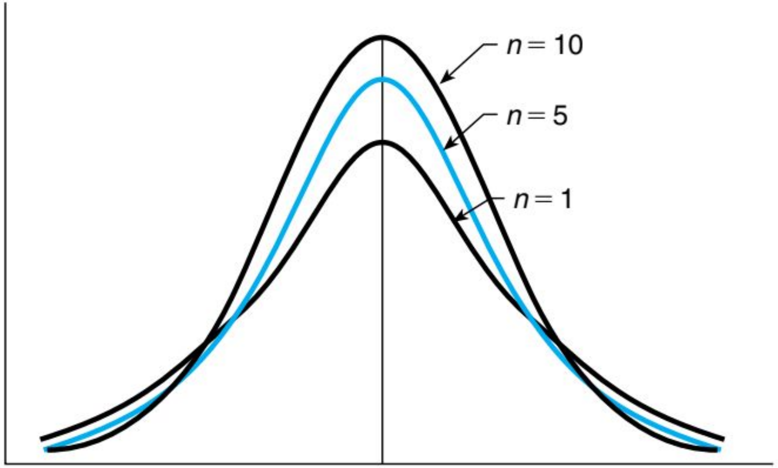


FIGURE 5.14 *Density function of T_n.*

Derivation of student-t distribution

- The PDF of a chi-square random variable with n degrees of freedom, divided by n is given as follows:

$$f_{\chi_n^2}(x) = \frac{x^{n/2-1}e^{-n/2}}{2^{n/2}\Gamma(n/2)}$$

$$Y = \chi_n^2/n \implies f_Y(x) = nf_{\chi_n^2}(nx) = \frac{n^{n/2}x^{n/2-1}e^{-n/2}}{2^{n/2}\Gamma(n/2)}$$

Incidentally, this is called a **Gamma random variable**.

Derivation of student-t distribution

$X \sim N(0,1)$ (standard normal),
$U \sim \chi^2_n$ (chi-square with n d.f.),
X and U independent,

$$T = \frac{X}{\sqrt{Z}}, \qquad \text{where } Z = \frac{U}{n}.$$

Equivalently $T = X\sqrt{\frac{n}{U}}$. We want the pdf $f_T(t)$.

1. Conditional distribution of T given $U = u$

Given $U = u$, $Z = u/n$ is fixed, so

$$T \mid U = u \sim N(0,\ n/u).$$

$$f_{T|U}(t \mid u) = \frac{1}{\sqrt{2\pi(n/u)}} \exp\Big(-\frac{t^2}{2(n/u)}\Big) = \sqrt{\frac{u}{2\pi n}}\, \exp\Big(-\frac{u\,t^2}{2n}\Big).$$

2. Density of U (chi-square)

The chi-square density for U is

$$f_U(u) = \frac{1}{2^{n/2}\,\Gamma(n/2)}\, u^{\frac{n}{2}-1}\, e^{-u/2}, \qquad u > 0.$$

3. Marginal density of T by integrating out U

$$\begin{aligned} f_T(t) &= \int_0^\infty f_{T|U}(t \mid u)\, f_U(u)\, du \\ &= \int_0^\infty \Big(\sqrt{\frac{u}{2\pi n}}\, e^{-ut^2/(2n)}\Big)\Big(\frac{1}{2^{n/2}\Gamma(n/2)}\, u^{\frac{n}{2}-1} e^{-u/2}\Big)\, du \\ &= \frac{1}{\sqrt{2\pi n}\; 2^{n/2}\Gamma(n/2)} \int_0^\infty u^{\frac{n+1}{2}-1}\, \exp\Big(-\frac{u}{2}\Big(1+\frac{t^2}{n}\Big)\Big)\, du. \end{aligned}$$

4. Evaluate the integral (Gamma integral)

Let $a = \frac{n+1}{2}$ and $b = \frac{1+t^2/n}{2}$. The integral is

$$\int_0^\infty u^{a-1} e^{-bu}\, du = \frac{\Gamma(a)}{b^a}.$$

So

$$f_T(t) = \frac{1}{\sqrt{2\pi n}\, 2^{n/2} \Gamma(n/2)} \cdot \frac{\Gamma\left(\frac{n+1}{2}\right)}{\left(\frac{1+t^2/n}{2}\right)^{\frac{n+1}{2}}}.$$

5. Simplify constants

$$\left(\frac{1+t^2/n}{2}\right)^{\frac{n+1}{2}} = \frac{(1+t^2/n)^{\frac{n+1}{2}}}{2^{\frac{n+1}{2}}},$$

so the prefactor simplifies because $2^{(n+1)/2}/2^{n/2} = 2^{1/2}$. Thus

$$\frac{2^{(n+1)/2}}{2^{n/2}\sqrt{2\pi n}} = \frac{\sqrt{2}}{\sqrt{2\pi n}} = \frac{1}{\sqrt{\pi n}}.$$

Putting everything together gives the standard Student-t density:

$$\boxed{f_T(t) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi}\,\Gamma\left(\frac{n}{2}\right)}\left(1+\frac{t^2}{n}\right)^{-\frac{n+1}{2}}, \qquad t \in \mathbb{R}.}$$

This turns out to be the beta function

(b) Hypothesis testing: mean of a normal distribution with unknown variance

- The statistic of interest here is given by T defined as follows

$$T = \frac{\sqrt{n}(\overline{X} - \mu_0)}{S}$$

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}$$

$$\overline{X} = \sum_{i=1}^{n} X_i / n$$

T follows a student-t distribution with *n*-1 degrees of freedom because S^2 is a chi-square random variable with *n*-1 degrees of freedom and it is independent of bar(X).

We have not yet proved this independence in our class - the proof requires multivariate statistics and is complicated.

Hypothesis testing: mean of a normal distribution with unknown variance

- Hence, we have the following:

$$P_{\mu_0}\left\{-t_{\alpha/2,\,n-1} \le \frac{\sqrt{n}(\overline{X}-\mu_0)}{S} \le t_{\alpha/2,\,n-1}\right\} = 1-\alpha$$

$t_{\alpha/2,n-1}$ is the 100 $\alpha/2$ upper percentile value of the t-distribution

$$\text{accept} \quad H_0 \quad \text{if} \quad \left|\frac{\sqrt{n}(\overline{X}-\mu_0)}{S}\right| \le t_{\alpha/2,\,n-1}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \left|\frac{\sqrt{n}(\overline{X}-\mu_0)}{S}\right| > t_{\alpha/2,\,n-1}$$

There are tables for the t distribution (just like there are for standard Gaussians) which give the value of $t_{\alpha/2,n-1}$ for any α.

Hypothesis testing: mean of a normal distribution with unknown variance

- The test on the previous slide is called the two-sided t test.
- Here let t be the value of the observed test statistic T.
- Then the p-value is the probability that $|T|$ exceeds $|t|$ if H_0 were to be true.
- The t-test then calls for rejection at all significance levels higher than the p-value and acceptance at all lower significance levels.

TABLE 8.2 *$X_1, \ldots, X_n$ Is a Sample from a $\mathcal{N}(\mu, \sigma^2)$ Population*

σ^2 Is Unknown, $\overline{X} = \sum_{i=1}^{n} X_i/n$ $S^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$

H_0	H_1	Test Statistic TS	Significance Level α Test	p-Value if $TS = t$
$\mu = \mu_0$	$\mu \neq \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $\lvert TS \rvert > t_{\alpha/2,\, n-1}$	$2P\{T_{n-1} \geq \lvert t \rvert\}$
$\mu \leq \mu_0$	$\mu > \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $TS > t_{\alpha,\, n-1}$	$P\{T_{n-1} \geq t\}$
$\mu \geq \mu_0$	$\mu < \mu_0$	$\sqrt{n}(\overline{X} - \mu_0)/S$	Reject if $TS < -t_{\alpha,\, n-1}$	$P\{T_{n-1} \leq t\}$

T_{n-1} is a t-random variable with $n-1$ degrees of freedom: $P\{T_{n-1} > t_{\alpha, n-1}\} = \alpha$.

Testing equality of means of two normal populations with known variances

- Consider $X_1, X_2, \ldots, X_n$ and $Y_1, Y_2, \ldots, Y_m$ are independent samples from two normal distributions with unknown means and known variances $(\sigma_x)^2$ and $(\sigma_y)^2$.
- The hypothesis to test here is given by:

$$H_0 : \mu_x = \mu_y \qquad\qquad H_1 : \mu_x \neq \mu_y$$

- We need to determine the distribution of the following quantity

$$\bar{X} - \bar{Y} \text{ where } \bar{X} = \sum_{i=1}^{n} X_i/n, \bar{Y} = \sum_{i=1}^{m} Y_i/m$$

$$\overline{X} - \overline{Y} \sim \mathcal{N}\left(\mu_x - \mu_y, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}\right)$$

$$\frac{\overline{X} - \overline{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim \mathcal{N}(0, 1)$$

$$(\overline{X} - \overline{Y}) \Bigg/ \sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}} \sim \mathcal{N}(0, 1)$$

If $\mu_x = \mu_y$

$$P_{H_0}\left\{-z_{\alpha/2} \leq \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \leq z_{\alpha/2}\right\} = 1 - \alpha$$

Testing equality of means of two normal populations with known variances

- The final test is given as follows (it is called a paired z-test)

$$\text{accept} \quad H_0 \quad \text{if} \quad \frac{|\overline{X} - \overline{Y}|}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}} \leq z_{\alpha/2}$$

$$\text{reject} \quad H_0 \quad \text{if} \quad \frac{|\overline{X} - \overline{Y}|}{\sqrt{\sigma_x^2/n + \sigma_y^2/m}} \geq z_{\alpha/2}$$

Testing equality of means of two normal populations with unknown but equal variances

- Consider X1, X2,...,Xn and Y1, Y2,..., Ym are independent samples from two normal distributions with **unknown** means μ1, μ2 and **unknown but equal** variances $(\sigma_1)^2 = (\sigma_2)^2 = \sigma^2$.
- The null and alternative hypotheses are as follows:

$$H_0 : \mu_x = \mu_y \qquad H_1 : \mu_x \neq \mu_y$$

- Again we would like to compute bar(X) and bar(Y) and reject H0 if bar(X)-bar(Y) is “far away from” 0.

$$\bar{X} - \bar{Y} \text{ where } \bar{X} = \sum_{i=1}^{n} X_i/n, \bar{Y} = \sum_{i=1}^{m} Y_i/m$$

Testing equality of means of two normal populations with unknown but equal variances

- To determine the meaning of “far away from zero”, we need the following:

$$S_1^2 = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{n-1} \qquad S_2^2 = \sum_{i=1}^{m} \frac{(Y_i - \overline{Y})^2}{m-1}$$

$$(n-1)\frac{S_1^2}{\sigma^2} \sim \chi^2_{n-1} \qquad (m-1)\frac{S_2^2}{\sigma^2} \sim \chi^2_{m-1}$$

$$(n-1)\frac{S_1^2}{\sigma^2} + (m-1)\frac{S_2^2}{\sigma^2} \sim \chi^2_{n+m-2}$$

Testing equality of means of two normal populations with unknown but equal variances

- What test statistic do we consider? To derive this, observe:

$$\overline{X} - \overline{Y} \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma^2}{n} + \frac{\sigma^2}{m}\right)$$

$$\frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma^2}{n} + \frac{\sigma^2}{m}}} \sim \mathcal{N}(0, 1)$$

$$S_p^2 = \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}$$

Testing equality of means of two normal populations with unknown but equal variances

- What test statistic do we consider? To derive this, observe:

$$\frac{\bar{X}-\bar{Y}-(\mu_1-\mu_2)}{\sqrt{\sigma^2(1/n+1/m)}} \div \sqrt{S_p^2/\sigma^2} = \frac{\bar{X}-\bar{Y}-(\mu_1-\mu_2)}{\sqrt{S_p^2(1/n+1/m)}}$$

Standard normal

Chi-square with *n*+*m*-2 degrees of freedom

t-distribution with *n*+*m*-2 degrees of freedom – note that bar(X), bar(Y), S_1, S_2 are all independent random variables. Hence bar(X), bar(Y) and S_p are also independent.

Testing equality of means of two normal populations with unknown but equal variances

$$P\left\{-t_{\alpha/2,n+m-2} \leq \frac{\overline{X}-\overline{Y}-(\mu_1-\mu_2)}{S_p\sqrt{1/n+1/m}} \leq t_{\alpha/2,n+m-2}\right\} = 1-\alpha$$

$$T \equiv \frac{\overline{X}-\overline{Y}}{\sqrt{S_p^2(1/n+1/m)}}$$

$$\begin{aligned} &\text{accept} \quad H_0 \quad \text{if} \quad |T| \leq t_{\alpha/2,\,n+m-2} \\ &\text{reject} \quad H_0 \quad \text{if} \quad |T| > t_{\alpha/2,\,n+m-2} \end{aligned}$$

Testing equality of means of two normal populations with unknown but unequal variances

- The test statistic will be computed using sample variances of X1,...,Xn and Y1,...,Ym.
- The hypotheses are:

$$H_0 : \mu_x = \mu_y \qquad \text{versus} \qquad H_1 : \mu_x \neq \mu_y$$

- The test statistic is:

$$\frac{\overline{X} - \overline{Y}}{\sqrt{\dfrac{S_x^2}{n} + \dfrac{S_y^2}{m}}}$$

The distribution of this test statistic is complicated and depends upon unknown parameters even when H0 is true. There is no known satisfactory solution to this problem.

Testing equality of means of two normal populations with unknown but unequal variances

- The test statistic is:

$$\frac{\overline{X} - \overline{Y}}{\sqrt{\dfrac{S_x^2}{n} + \dfrac{S_y^2}{m}}}$$

The distribution of this test statistic is complicated and depends upon unknown parameters even when H0 is true. There is no known satisfactory solution to this problem.

- The only resort is when n and m are very large – it can be shown that when H0 is true, then the statistic has a standard normal distribution.

Testing equality of means of two normal populations with unknown but unequal variances

- The corresponding test is as follows:

$$\text{accept} \quad H_0 \quad \text{if} \quad -z_{\alpha/2} \leq \frac{\overline{X} - \overline{Y}}{\sqrt{\dfrac{S_x^2}{n} + \dfrac{S_y^2}{m}}} \leq z_{\alpha/2}$$

$$\text{reject} \quad \text{otherwise}$$

Hypothesis test regarding the variance of a normal population (with unknown mean)

- Let X1,...,Xn be independent samples from a normal distribution with unknown mean μ and unknown variance σ^2.
- The hypothesis of interest is given as follows:

$$H_0 : \sigma^2 = \sigma_0^2 \qquad H_1 : \sigma^2 \neq \sigma_0^2$$

- We compute the sample variance:

$$S_x^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{n-1}$$

We have seen that (n-1)$(S_x)^2/\sigma^2$ is a chi-square random variable with n-1 degrees of freedom.

Hypothesis test regarding the variance of a normal population (with unknown mean)

- Hence when H0 is true, we have

$$\frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2$$

$$P\{\chi_{n-1}^2 < \chi_{\alpha/2,n-1}^2\} = 1 - \alpha/2 \qquad P\{\chi_{n-1}^2 < \chi_{1-\alpha/2,n-1}^2\} = \alpha/2$$

$$P_{H_0}\left\{\chi_{1-\alpha/2,n-1}^2 \le \frac{(n-1)S^2}{\sigma_0^2} \le \chi_{\alpha/2,n-1}^2\right\} = 1 - \alpha$$

Hypothesis test regarding the variance of a normal population (with unknown mean)

- The hypothesis test is:

$$\begin{array}{llll} \text{accept} & H_0 & \text{if} & \chi^2_{1-\alpha/2,n-1} \le \dfrac{(n-1)S^2}{\sigma_0^2} \le \chi^2_{\alpha/2,n-1} \\ \text{reject} & H_0 & \text{otherwise} & \end{array}$$

- Compute the test statistic value c from the samples.
- Compute the probability P1 that a chi-square random variable with n-1 degrees of freedom would be less than c.
- Compute the probability P2 that a chi-square random variable with n-1 degrees of freedom would be greater than c.
- If either probability is less than $\alpha/2$, the hypothesis H0 is rejected.

$$p\text{-value} = 2\min(P\{\chi^2_{n-1} < c\}, 1 - P\{\chi^2_{n-1} < c\})$$

Hypothesis test for the success parameter of a Bernoulli distribution

- Consider *n* independent Bernoulli trials with success parameter *p*.
- The number of heads/successes X in *n* trials has a binomial distribution with parameters (*n*,*p*).
- The null and alternate hypotheses here are as follows, given a pre-specified parameter value p_0:

$$H_0 : p \leq p_0 \quad \text{versus} \quad H_1 : p > p_0$$

- We will reject H_0 if X is “too large”.
- This sort of a test is useful if X denotes the number of defective items in a set of *n* presented items.
- How large should X be in order to justify rejection at significance level α?

Hypothesis test for the success parameter of a Bernoulli distribution

- Note the following tail probability:

$$P\{X \geq k\} = \sum_{i=k}^{n} P\{X = i\} = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

- Intuitively, $P(X \geqslant k)$ is an increasing function of p.
- When H0 is true, i.e. when we have $p \leqslant p0$, we also get the following bound:

$$P\{X \geq k\} \leq \sum_{i=k}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i}$$

Hypothesis test for the success parameter of a Bernoulli distribution

- Hence a significance level α test for H0: $p \leqslant p0$ versus H1: $p > p0$ implies rejecting H0 when $X \geqslant k^*$ where k^* is defined as follows:

$$k^* = \min\left\{k : \sum_{i=k}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i} \leq \alpha\right\}$$

k^* is the smallest value for which:

$$\sum_{i=k}^{n} \binom{n}{i} p_0^i (1-p_0)^{n-i} \leq \alpha$$

Hypothesis test for the success parameter of a Bernoulli distribution

- The associated p-value is as follows:

$$
\begin{aligned}
p\text{-value} &= P\{B(n, p_0) \geq x\} \\
&= \sum_{i=x}^{n} \binom{n}{i} p_0^i (1 - p_0)^{n-i}
\end{aligned}
$$

Hypothesis test: equality of success parameters in two Bernoulli distributions

- Consider two methods - method **M1** and method **M2** - to manufacture a device.
- Let the probability of obtain defective devices using **M1** be p_1.
- Let the probability of obtain defective devices using **M2** be p_2.
- We want to test the hypothesis $p_1 = p_2$, for which we will produce n_1 devices using method **M1** and n_2 devices using method **M2**.

$$H_0 : p_1 = p_2 \quad \text{versus} \quad H_1 : p_1 \neq p_2$$

- Let X_1 = number of defective devices using **M1**, X_2 = number of defective devices using **M2**
- Now X_1 ~ Binomial(n1,p1) and X_2 ~Binomial(n_2,p_2).
- Let k be the number of defectives, that is $X_1+X_2=k$.
- How do we check whether $p_1 = p_2$?

Hypothesis test: equality of success parameters in two Bernoulli distributions

- If indeed $p_1 = p_2$ (that is H0 is true), then each of the n_1+n_2 devices will have the same probability of being defective.
- In that case, the number of defectives (*k*) will have the same distribution as a random selection of a sample of k devices from a populations of n_1 items of type M1 and n_2 items of type M2.
- This is given as follows:

$$P_{H_0}\{X_1 = i | X_1 + X_2 = k\} = \frac{\binom{n_1}{i}\binom{n_2}{k-i}}{\binom{n_1+n_2}{k}}, \quad i = 0, 1, \ldots, k$$

Hypothesis test: equality of success parameters in two Bernoulli distributions

- It is reasonable to reject H0 if the proportion of defective items by methods **M1** and **M2** are very different.
- That is we reject H0 if the ratio X_1/n_1 is very different from $X_2/n_2 = (k-X_1)/n_2$.
- These ratios are very different is X_1 is either "too small", i.e. close to 0, or "too large", i.e. close to k.

Hypothesis test: equality of success parameters in two Bernoulli distributions

- At significance level α, this leads to the following where X is a hypergeometric random variable (X here plays the role of X_1 on slide 56):

$$\begin{array}{lll} \text{reject} & H_0 & \text{if either} \quad P\{X \leq x_1\} \leq \alpha/2 \quad \text{or} \quad P\{X \geq x_1\} \leq \alpha/2 \\ \text{accept} & H_0 & \text{otherwise} \end{array}$$

$$P\{X = i\} = \frac{\binom{n_1}{i}\binom{n_2}{k-i}}{\binom{n_1+n_2}{k}} \quad i = 0, 1, \ldots, k$$

$$p\text{-value} = 2\min(P\{X \leq x_1\}, P\{X \geq x_1\})$$

(1) Example application of hypothesis testing

- Among a clinic's patients having blood cholesterol levels ranging in the medium to high range (at least 220 milliliters per deciliter of serum), volunteers were recruited to test a new drug designed to reduce blood cholesterol.
- A group of **50** volunteers was given the drug for 1 month and the changes in their blood cholesterol levels were noted.
- If the average change was a reduction of **14.8** with a sample standard deviation of **6.4**, what conclusions can be drawn?
- **Solution**: We need to test whether the changes in cholesterol was solely due to chance.
- Sample mean reduction = bar(X) = 14.8
- Sample standard deviation of reduction = 6.4
- Value of test statistic for a t-test (why t-test here?) is

$$T = \sqrt{n}\,\overline{X}/S = \sqrt{50}\,14.8/6.4 = 16.352$$

As per the table, we should clearly reject the hypothesis that this was purely chance.

(1) Example application of hypothesis testing

- Despite this, we cannot fully reject the possibility of chance. Why?
- There could be a placebo effect: the medicines received can improve a patient's health due to psychological reasons
- The weather conditions during consumption of the new medicine could have had an effect.
- A better experiment is one that falsifies other possibilities for the reduction of blood cholesterol.
- Divide the volunteers into two (roughly equal) groups at **random**:
  - One group receives the drug
  - The other group receives the placebo (control group) – which is designed to look and taste similar to the actual drug
  - Neither group is told which one they are receiving
  - Any difference in performance between the two groups tells us whether the new drug is actually working

(2) Example application of hypothesis testing

- **22** volunteers at a research institute caught a cold after having been exposed to various cold viruses.
- A random selection of **10** of these volunteers was given tablets containing 1 gram of vitamin C. These tablets were taken four times a day.
- The control group consisting of the other **12** volunteers was given placebo tablets that looked and tasted exactly the same as the vitamin C tablets.
- This was continued for each volunteer until a doctor, who did not know if the volunteer was receiving the vitamin C or the placebo tablets, decided that the volunteer was no longer suffering from the cold. The length of time the cold lasted was then recorded and is given on the next slide.

(2) Example application of hypothesis testing

Treated with Vitamin C	Treated with Placebo
5.5	6.5
6.0	6.0
7.0	8.5
6.0	7.0
7.5	6.5
6.0	8.0
7.5	7.5
5.5	6.5
7.0	7.5
6.5	6.0
	8.5
	7.0

Do the data listed prove that taking 4 grams daily of vitamin C reduces the mean length of time a cold lasts? At what level of significance?

(2) Example application of hypothesis testing

We need to reject the null hypothesis that the mean time of cold when the placebo was taken (μ_P) is less than or equal to when vitamin C was taken (μ_C).

$$H_0 : \mu_p \leq \mu_c \qquad \text{versus} \qquad H_1 : \mu_p > \mu_c$$

$$\overline{X} = 6.450, \qquad \overline{Y} = 7.125$$

$$S_x^2 = .581, \qquad S_y^2 = .778$$

X: vitamin C
Y: placebo

$$S_p^2 = \frac{9}{20} S_x^2 + \frac{11}{20} S_y^2 = .689$$

$$TS = \frac{-.675}{\sqrt{.689(1/10 + 1/12)}} = -1.90$$

$$t_{.05,20} = 1.725$$

H0 is rejected at the 5 percent significance level.
P-value is around 0.03.

Goodness of Fit Tests

- Suppose we are given n iid samples $Y_1, Y_2,..., Y_n$.
- Consider a distribution F(y).
- We want to ask the question: Are these samples from this particular distribution?
- Note that merely evaluating the likelihood and thresholding it is inadequate – the threshold won't have a specific meaning unlike what we did in hypothesis testing so far.
- Statistical tests that determine whether or not the samples came from the specified distribution are called **goodness of fit** tests.

Goodness of Fit Tests: Discrete case

- Consider n iid samples $Y_1, Y_2, \ldots, Y_n$ where each takes on values from 1,2...,*k*.
- We want to determine whether these samples came from a distribution given by $\{p_i\}$ for i=1 to k.
- Here the null (H0) and alternative (H1) hypotheses are given as follows where Y denotes any of the samples Yi.

$$H_0 : P\{Y = i\} = p_i, \qquad i = 1, \ldots, k$$

$$H_1 : P\{Y = i\} \neq p_i, \qquad \text{for some } i = 1, \ldots, k$$

Goodness of Fit Tests: Discrete case

- To test the hypothesis, we determine the number X_i of samples that are equal to *i* where *i* lies from 1 to *k*.
- A sample Y_j equals i with probability P(Y=i) = p_i. Under null hypothesis, X_i will be a binomial random variable with parameters n and p_i.
- When H0 is true, we have $E(X_i) = np_i$.
- The squared difference $(X_i - np_i)^2$ indicates something about how likely it is that p_i equals P(Y=*i*).
- When this difference is very large as compared to np_i, then we would be inclined to believe that the null hypothesis is to be rejected.
- The test statistic is as follows:

$$T = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i}$$

Goodness of Fit Tests: Discrete case

- If *n* is large, then the distribution of T will asymptotically be **chi-square** with *k*-1 degrees of freedom.
- Remember: *n* is the number of samples, *k* = number of possible values of Y.
- There are *k*-1 and not *k* degrees of freedom, because one degree of freedom is lost because the X_i values sum up to *n*, that is: $\sum_i X_i = n$
- The proof about the distribution of T is complicated, but we will show it for *k*=2:

$$
\begin{aligned}
T &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2} \\
&= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1 - p_1])^2}{n(1 - p_1)}
\end{aligned}
$$

$$
\begin{aligned}
X_1 + X_2 &= n \\
p_1 + p_2 &= 1
\end{aligned}
$$

$$= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1-p_1)}$$

$$= \frac{(X_1 - np_1)^2}{np_1(1-p_1)} \qquad \text{since} \qquad \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}$$

- X1 ~ Binomial (n,p_1)
- But when n is large, then $X_1 - np_1$ is also a zero-mean Gaussian with variance $np_1(1-p_1)$ – by the **Central Limit Theorem** or the de-Moivre Laplace theorem.
- Hence the term on the RHS of the last equation above is approximately a **standard normal**.
- Hence T is approximately a **chi-square** distribution with k-1=1 degree of freedom.
- What about the case for arbitrary k? We will prove it when we do multivariate Gaussian distribution (our next chapter).

Goodness of Fit Tests: Discrete case

- The hypothesis test is now expressed as follows:

$$\begin{array}{lll} \text{reject} & H_0 & \text{if} \quad T \geq \chi^2_{\alpha,k-1} \\ \text{accept} & H_0 & \text{otherwise} \end{array}$$

$$\begin{aligned} p\text{-value} &= P_{H_0}\{T \geq t\} \\ &\approx P\{\chi^2_{k-1} \geq t\} \end{aligned}$$

- This is called the **chi-square goodness of fit test**.
- Important point to note: The distribution of T is completely **independent of the underlying PMF** given the null hypothesis!
- So even if the PMF changes, the distribution of T remains the **same.**

Goodness of fit tests: continuous case

- Consider n iid **continuous-valued** samples $Y_1, Y_2, \dots, Y_n$
- We want to test the hypothesis H_0 that the samples are from the continuous distribution F(y).
- For this, we can perform **discretization** and split the continuous values for the Y's into some k non-overlapping intervals given as:

$$(y_0, y_1), (y_1, y_2), \ldots, (y_{k-1}, y_k), \qquad \text{where} \quad y_0 = -\infty, y_k = +\infty$$

- We will get **discretized** versions of the original samples

$$Y_j^d = i \qquad \text{if } Y_j \text{ lies in the interval } (y_{i-1}, y_i)$$

Goodness of fit tests: continuous case

- If the null hypothesis is true, then the samples are indeed from F(y).
- In such a case, we will have:

$$P\{Y_j^d = i\} = F(y_i) - F(y_{i-1}), \qquad i = 1, \ldots, k$$

- In such a case, we can use the chi-square test for goodness of fit.
- How discretization is **expensive computationally** (if k is chosen to be large) and results in **loss of accuracy**.
- Can we have a test which respects continuous nature of the samples and does not use discretization?

Goodness of fit tests: continuous case

- We resort to the use of the empirical CDF defined as follows:

$$F_e(x) = \frac{\#i : Y_i \leq x}{n}$$

- Note that there is no discretization here.
- The empirical CDF is an **unbiased** and **consistent** estimator for the CDF.
- In other words, we have:

$$E(F_e(x)) = F(x) \quad \mathrm{Var}(F_e(x)) = \frac{1}{n}F(x)(1 - F(x))$$

$$E(F_e(x)) = E\left(\frac{1}{n}\sum_{i=1}^{n}\mathcal{I}(Y_i \leq x)\right)$$

$$= \frac{1}{n}\sum_{i=1}^{n}E(\mathcal{I}(Y_i \leq x))$$

$$= \frac{1}{n}\sum_{i=1}^{n}P(Y_i \leq x)$$

$$= \frac{1}{n}\sum_{i=1}^{n}F(x) = F(x)$$

Indicator function - gives you 1 if the predicate is true, otherwise it gives you zero

A Bernoulli r.v. with success probability P(Y_i <= x)

The empirical CDF is an **unbiased** estimator of the true CDF!

$$\mathrm{Var}(F_e(x)) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n}\mathcal{I}(Y_i \leq x)\right)$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(\mathcal{I}(Y_i \leq x))$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}P(Y \leq x)(1 - P(Y \leq x))$$

$$= \frac{1}{n}F(x)(1 - F(x))$$

The variance of the empirical CDF drops to 0 as n increases. As the bias is 0, this means that the MSE also drops to 0 as n increases.

Goodness of fit tests: continuous case

- If the null hypothesis is true, we will see that $F_e(x)$ is "close to" $F(x)$ for any x.
- So we consider the following test statistic – called the Komogorov Smirnov Test statistic – where x ranges from -infinity to +infinity:

$$D \equiv \underset{x}{\text{Maximum}} \, |F_e(x) - F(x)|$$

- The surprising thing is that the distribution of D does not depend on F(x) when the null hypothesis is true!
- We will prove this – you already did it in your homework :-)

$$P_F\{D \geq d\} = P_F\left\{\underset{x}{\text{Maximum}}\left|\frac{\#i : Y_i \leq x}{n} - F(x)\right| \geq d\right\}$$

$$= P_F\left\{\underset{x}{\text{Maximum}}\left|\frac{\#i : F(Y_i) \leq F(x)}{n} - F(x)\right| \geq d\right\}$$

$$= P\left\{\underset{x}{\text{Maximum}}\left|\frac{\#i : U_i \leq F(x)}{n} - F(x)\right| \geq d\right\}$$

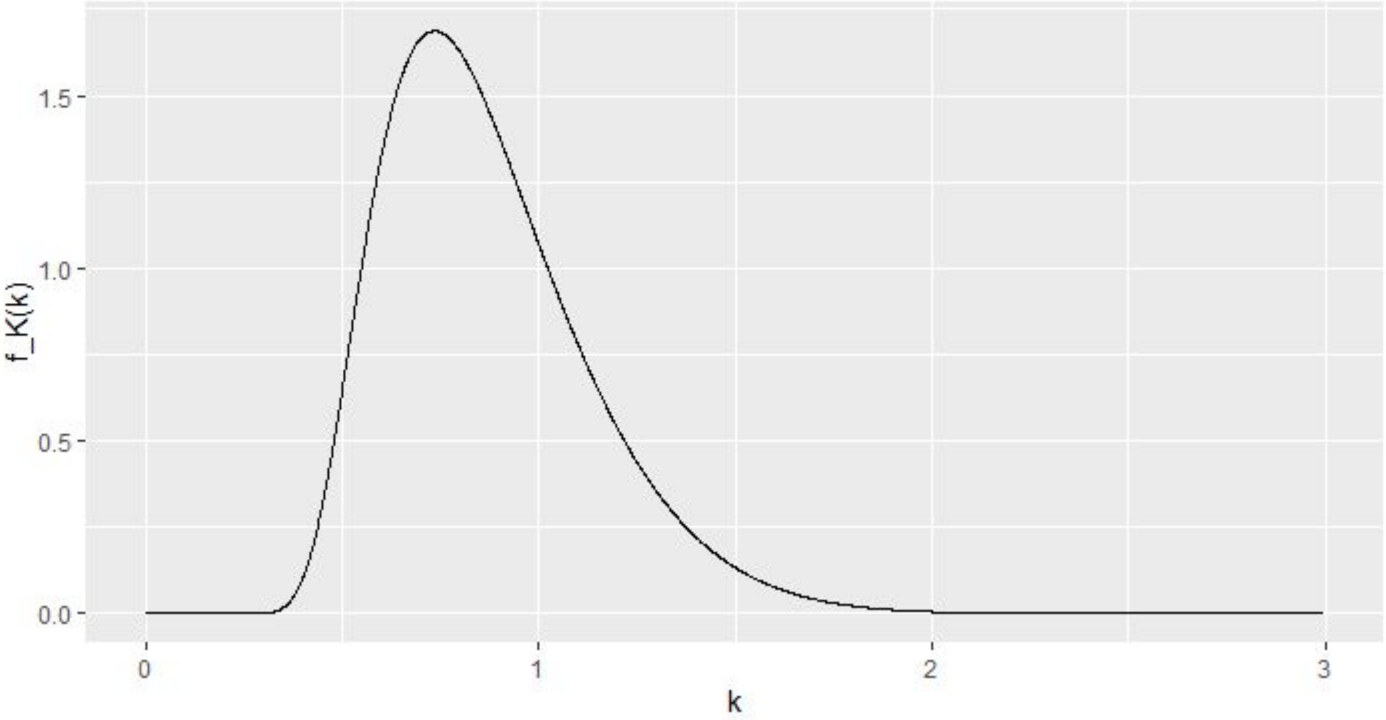
$$P_F\{D \geq d\} = P\left\{\underset{0 \leq y \leq 1}{\text{Maximum}}\left|\frac{\#i : U_i \leq y}{n} - y\right| \geq d\right\}$$

- Here each Ui is a Uniform(0,1) random variable.
- As x ranges from -infinity to +infinity, y=F(x) ranges from 0 to 1.
- From the above derivation, it is clear that the distribution of D does not depend on F when the null hypothesis is true.

Goodness of fit tests: continuous case

- We have established that the distribution of D is independent of F(x).
- We have not derived the distribution of D (it is called the Kolmogorov distribution), and it is indeed complicated.
- We will only state the formula here:

$$P(D \leq x) = \frac{\sqrt{2\pi}}{x} \sum_{k=1}^{\infty} e^{-(2k-1)^2\pi^2/(8x^2)}$$



https://en.wikipedia.org/wiki/Kolmogorov%E2%80%93Smirnov_test

$$P_F\{D^* \geq d^*_\alpha\} = \alpha$$

$$d^*_{.1} = 1.224, \qquad d^*_{.05} = 1.358, \qquad d^*_{.025} = 1.480, \qquad d^*_{.01} = 1.626$$

Goodness of fit tests: continuous case

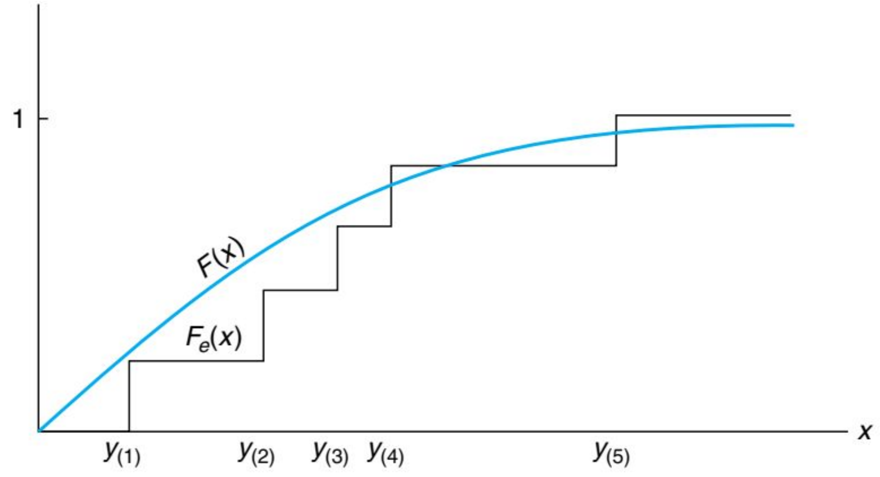
- How do we compute D efficiently?
- For this, the values of {Yi} are sorted in increasing order to yield the following:

$$y_{(j)} = j\text{th smallest of } y_1, \ldots, y_n$$

$$F_e(x) = \begin{cases} 0 & \text{if } x < y_{(1)} \\ \dfrac{1}{n} & \text{if } y_{(1)} \le x < y_{(2)} \\ \vdots & \\ \dfrac{j}{n} & \text{if } y_{(j)} \le x < y_{(j+1)} \\ \vdots & \\ 1 & \text{if } y_{(n)} \le x \end{cases}$$

Goodness of fit tests: continuous case

- $F_e(x)$ is a **piece-wise constant** function (like a staircase) - constant within $y_{(j-1)}$ to $y_{(j)}$ for any j=2 to *n*.
- It jumps by a factor of 1/n at each $y_{(j)}$.
- F(x) is strictly increasing.
- Hence the maximum value of $F_e(x)$-F(x) occurs at one of the sample values, i.e. at some $y_{(j)}$ for j = 1 to n.
- Also the maximum value of F(x)-Fe(x) occurs just before one of the sample values $y_{(j)}$ for j=1 to n.
- Due to this, the value of D can be efficiently computed in O(n) time.



Goodness of fit tests: continuous case

$$\underset{x}{\text{Maximum}}\{F(x) - F_e(x)\} = \underset{j=1,\ldots,n}{\text{Maximum}}\left\{F(y_{(j)}) - \frac{j-1}{n}\right\}$$

$$\underset{x}{\text{Maximum}}\{F_e(x) - F(x)\} = \underset{j=1,\ldots,n}{\text{Maximum}}\left\{\frac{j}{n} - F(y_{(j)})\right\}$$

$$\begin{aligned}
D &= \underset{x}{\text{Maximum}}\,|F_e(x) - F(x)| \\
&= \text{Maximum}\{\text{Maximum}\{F_e(x) - F(x)\}, \text{Maximum}\{F(x) - F_e(x)\}\} \\
&= \text{Maximum}\left\{\frac{j}{n} - F(y_{(j)}), F(y_{(j)}) - \frac{j-1}{n}, j = 1, \ldots, n\right\}
\end{aligned}$$

Digression: Dvoretzky-Kiefer-Wolfowitz inequality

- The empirical CDF is an excellent quality estimator of the true CDF.
- The MSE analysis a few slides ago gives you only the expectation of $(F_e(x)-F(x))^2$ at some value x.
- There is another interesting result about the maximum absolute deviation between $F_e(x)$ and $F(x)$ – with high probability.
- It is called the **Dvoretzky-Kiefer-Wolfowitz inequality** and it states that:

$$P\left(\sup_x |F_n(x) - F(x)| > \varepsilon\right) \leq 2e^{-2n\varepsilon^2}$$

  - The probability that the maximum deviation between $F_e(x)$ and $F(x)$ is greater than ϵ is very small, less than $\mathbf{exp(-n\epsilon^2)}$.
  - It tells you how powerful the Kolmogorov statistic is.
  - We won't derive this result in our class.